Subject:- Object Oriented Programming Through Java

Lab Internal Examination Programs

List Simple Program

1. Write a java program to print the sum of all elements of an array.
2. Write a java program to print Fibonacci series up to given number using recursion.
3. Write a java program to implement method overloading.
4. Write a java program to implement constructor overloading.
5. Write a program to demonstrate the following i) Single inheritance ii) Multi–level inheritance
6. Write a program to demonstrate the following (i) Super, ii) Final keywords
7. Write a program to demonstrate the usage of method overriding, calling super class constructor in derived class.
8. Write a Java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle, and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area () that prints the area of the given shape.
9. Write a java program to create two thread Odd and Even which will print odd and even number up to n respectively with delayed 1000 ms.
10. Write a java program to create user define exception class Negative which will throw the exception if entered number is Negative.
11. Write a java program that handles all mouse events and shows the event name at the centre of the window when a mouse event is fired (Use Adapter classes).
12. Write a program to demonstrate polymorphism using interface (interface in package P1and class in package P2)
13. Write a java program to list all the files in a directory including the files present in all its sub directories.
14. Write java program to demonstrate the JTree.
15. Write java program to demonstrate the JTable.
16. Write java program to demonstrate Arraylist and its methods.

List of Complex Programs

1. Write a Java program that implements a multi-thread application that has three threads. First thread generates a random integer every 1 second and if the value is even, these cond thread computes the square of the number and prints. If the value is odd, the third thread will print the value of the cube of the number.
2. Write a Java program that correctly implements the producer – consumer problem using the concept of inter thread communication
3. Write a Java program that creates a user interface to perform integer divisions. The user enters two numbers in the text fields, Num1 and Num2. The division of Num1 and Num 2 is displayed in the Result field when the Divide button is clicked. If Num1 or Num2 were not an integer, the program would throw a Number Format Exception. If Num2 were Zero, the program would throw an Arithmetic Exception. Display the exception in a message dialog box.
4. Write a Java program that works as a simple calculator. Use a grid layout to arrange buttons for the digits and for the +, -,*, % operations. Add a text field to display the result. Handle any possible exceptions like divided by zero.
5. Write a Java program that simulates a traffic light. The program lets the user select one of three lights: red, yellow, or green with radio buttons. On selecting a button, an appropriate message with “Stop” or “Ready” or “Go” should appear above the buttons in the selected color. Initially, there is no message shown.
6. Write a Java program for the following: Create a doubly linked list of elements. Delete a given element from the above list. Display the contents of the list after deletion.
7. Suppose that a table named Table.txt is stored in a text file. The first line in the file is the header, and the remaining lines correspond to rows in the table. The elements are separated by commas. Write a java program to display the table using Labels in Grid Layout.
8. Write a Java program that loads names and phone numbers from a text file where the data is organized as one line per record and each field in are cord are separated by a tab(\t). It takes a name or phone number as input and prints the corresponding other value from the hash table (hint: use hash tables).
9. Write a java program to find result of a given arithmetic expression? (EX: if you given arithmetic expression like 10+20-24*4/2-4.5= it should print 7.5)
10. Write java program to demonstrate the JList and display the selected item on the JLabel
11. Write a java program to create JTabbedPane which having three-tab Red, Green and Blue by clicking on each tab the panel will change the its background color respectively
12. Write a java program to create PopupMenu Cut, Copy and Paste with its short cut key Ctr+x, Ctr+c and Ctr+v respectively
13. Write a java program to create JComboBox having value 10,20,30,40,50 and apply the size on JLabel text “Welcome To OOPs-Lab” after selecting item from JComboBox
14. Write a Java program to demonstrate the order of constructor execution in a multilevel inheritance hierarchy.

15. Write a Java program to demonstrate the use of toString(), equals(), getClass() and clone () methods inherited from the Object class.
16. Write a Java program to demonstrate interface inheritance by extending one interface with another and implementing the extended interface in a class.